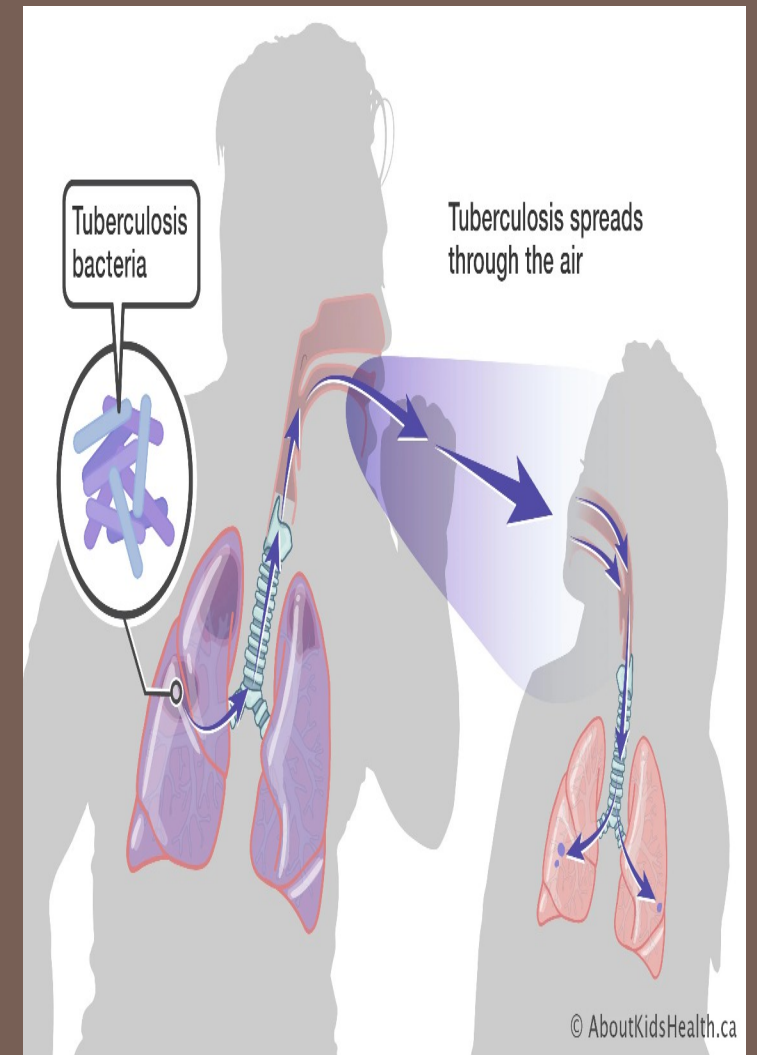




TUBERCULOSIS IN CHILDREN

09/13/2026

BY:- Dr. ACHALU M.

Outlines

2

- Learning objectives
- Epidemiology
- Etiology
- Transmission
- Pathogenesis
- Clinical features
- Diagnosis
- Treatment

Objectives

3

- Define bacteriologic characteristics & epidemiology of tuberculosis
- Discuss pathogenesis and clinical presentations of tuberculosis
- Outline different clinical forms of Tuberculosis
- Discuss on algorithm of TB diagnosis
- Demonstrate how to do and interpret TST
- State Management principles of tuberculosis
- Articulate the indications of steroid

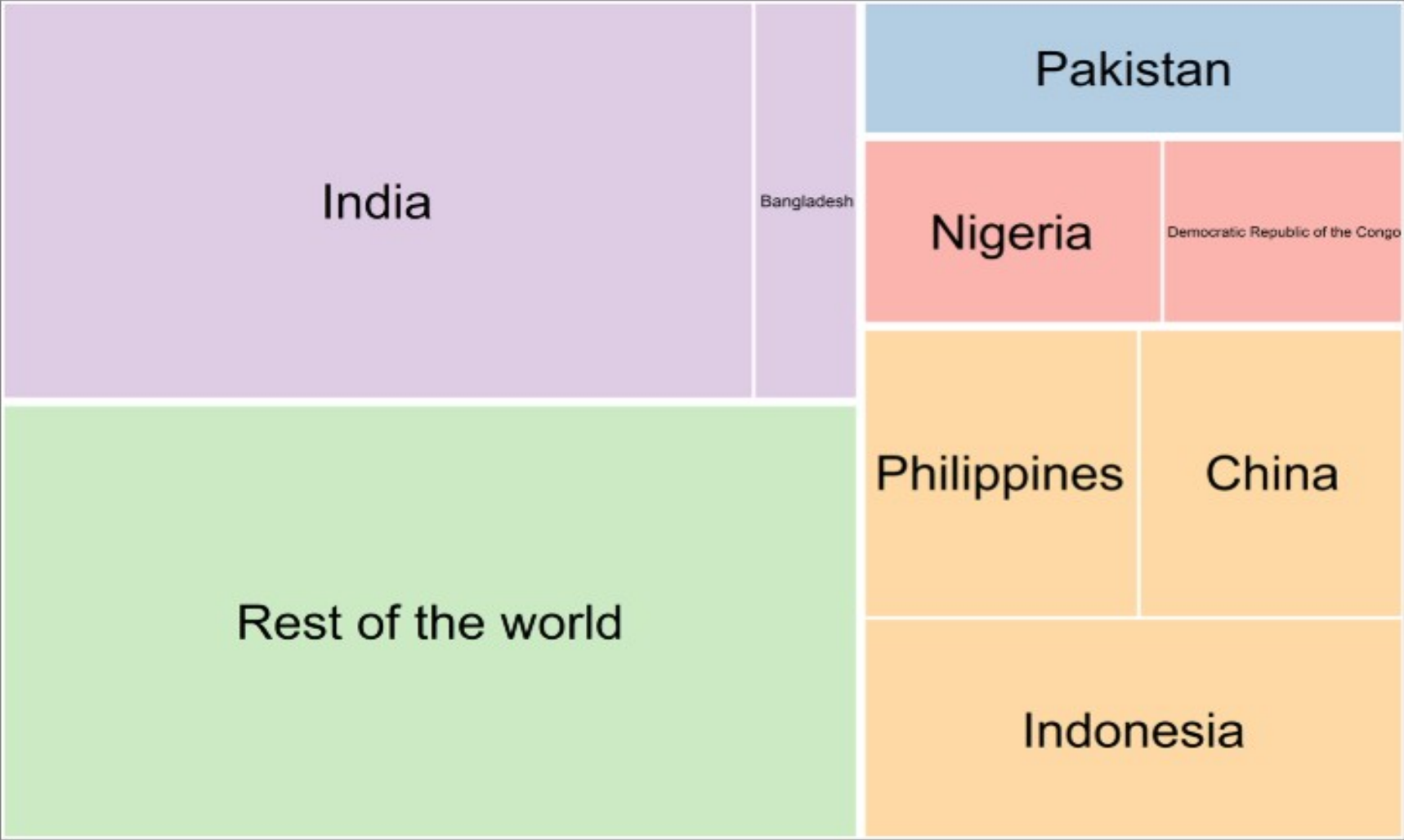
Epidemiology

4

- In 2024
 - $\approx$ 10.7m TB cases
 - 5.8m men
 - 3.7mwomen
 - 1.2mchildren
 - 1.23mdied from TB
 - 174,000.....<15yr died14% of death
- 8 countries account for 2/3rd cases
 - **India** is leading
 - Others **Indonesia, China, the Philippines, Pakistan, Nigeria, Bangladesh and D.R. CONGO**
- Ethiopia is 12th highly burdened country

Geographic distribution of the estimated number of incident TB cases in 2024

6



Four countriesaccounting 50% of MDR/RR-TB

9

- in **2024**:
 - India (32% of global cases),
 - China (7.1%),
 - Philippines (7.1%) and
 - Russian Federation (6.7%)

Epidemiology Africa and Ethiopia

12

- Africa bears a disproportionate burden of TB cases.
- **Ethiopia:** Ranks among the top 30 high TB burden countries globally.
- **Incidence rate in Ethiopia:** Approximately 92–124 cases per 100,000 population (fluctuates by year).

Pediatric TB vs. Adult TB

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- **Adults:** Often cavitary disease, high bacterial load, highly infectious.
- **Children:**
 - Paucibacillary (few bacteria), making diagnosis difficult.
 - Higher risk of progression from infection to disease.
 - Higher risk of disseminated disease (e.g., Miliary TB, TB Meningitis).
 - Children are rarely contagious; they usually acquire it from adults.

Etiology.

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- There are **5** closely related mycobacteria tuberculosis complex
 - M. tuberculosis..... Most common
 - M. bovis Rarely, via unpasteurized milk.
 - M. africanummainly In West Africa
 - M. microti..... Rarely affect humans
 - M. canetti.rare , limited human to human transmittion

Etiology.....

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- The tubercle bacilli are:
 - non-spore-forming
 - non-motile
 - pleomorphic
 - obligate aerobes
- Acid-fast bacilli (AFB) on Ziehl-Neelsen stain.
- Slow growing (doubling time 24 hours).

Ziehl-Neelsen stain.

16

	acid-fast bacteria	non-acid-fast bacteria
1 Before to acid-fast staining	 transparent	 transparent
2 After carbol fuchsin stain (primary stain of acid-fast staining)	 fuchsia	 fuchsia
3 After acid-alcohol (the decolorizer of acid-fast staining)	 fuchsia	 transparent
4 After methylene blue stain (the counterstain of acid-fast staining)	 fuchsia	 blue/purple

TB..... culture

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- **Solid media:-** Lowenstein-Jensen (LJ), Middlebrook media (7H10, 7H11)
 - Isolation from clinical specimens usually takes 3–6 wk &
 - Drug susceptibility testing requires additional 2- 4 wk.
- **Liquid media:-** MGIT (Mycobacteria Growth Indicator Tube), BACTEC
 - Growth can be detected in 1–3 wk
 - Drug susceptibilities can be determined in 3–5 days.

Etiology..... NAAT

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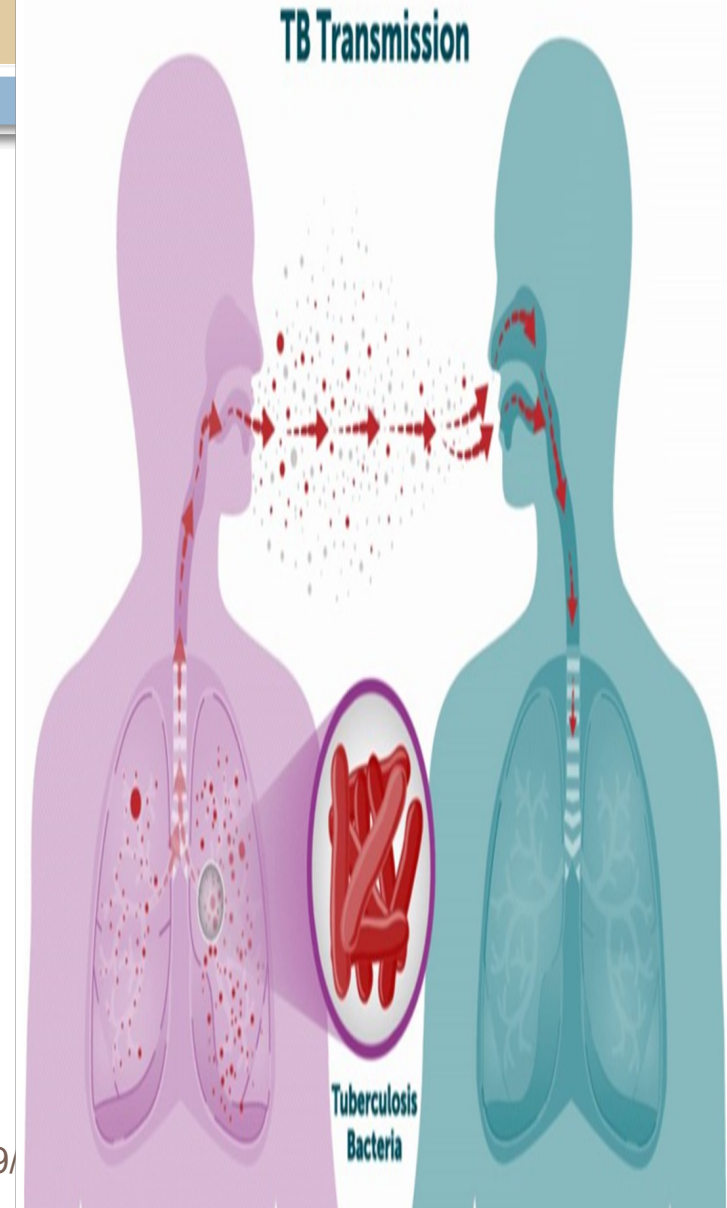
□ **Nucleic acid amplification (NAA) tests:**

- M. tuberculosis in clinical specimens sometimes can be detected directly within **hours**.
- Gene xpert
- Line probe assay
 - 24-48hr
 - Detect:- MTB, RR,HR
 - 2nd line fluoroquinolones, injectablesnewer LPA

Transmission.

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- Transmission of *M. tuberculosis* is:
 - person to person
 - usually by airborne mucus droplet nuclei
- Transmission **rarely** occurs by **direct**
 - Infected discharge or
 - Contaminated fomite.



Transmission.

20

- The chance of transmission ↑es when the patient has:
 1. Positive acid-fast smear of sputum
 2. extensive upper lobe infiltrate or cavity
 3. copious production of thin sputum
 4. severe and forceful cough.
 5. Environmental factors, especially poor air circulation

Transmission.....

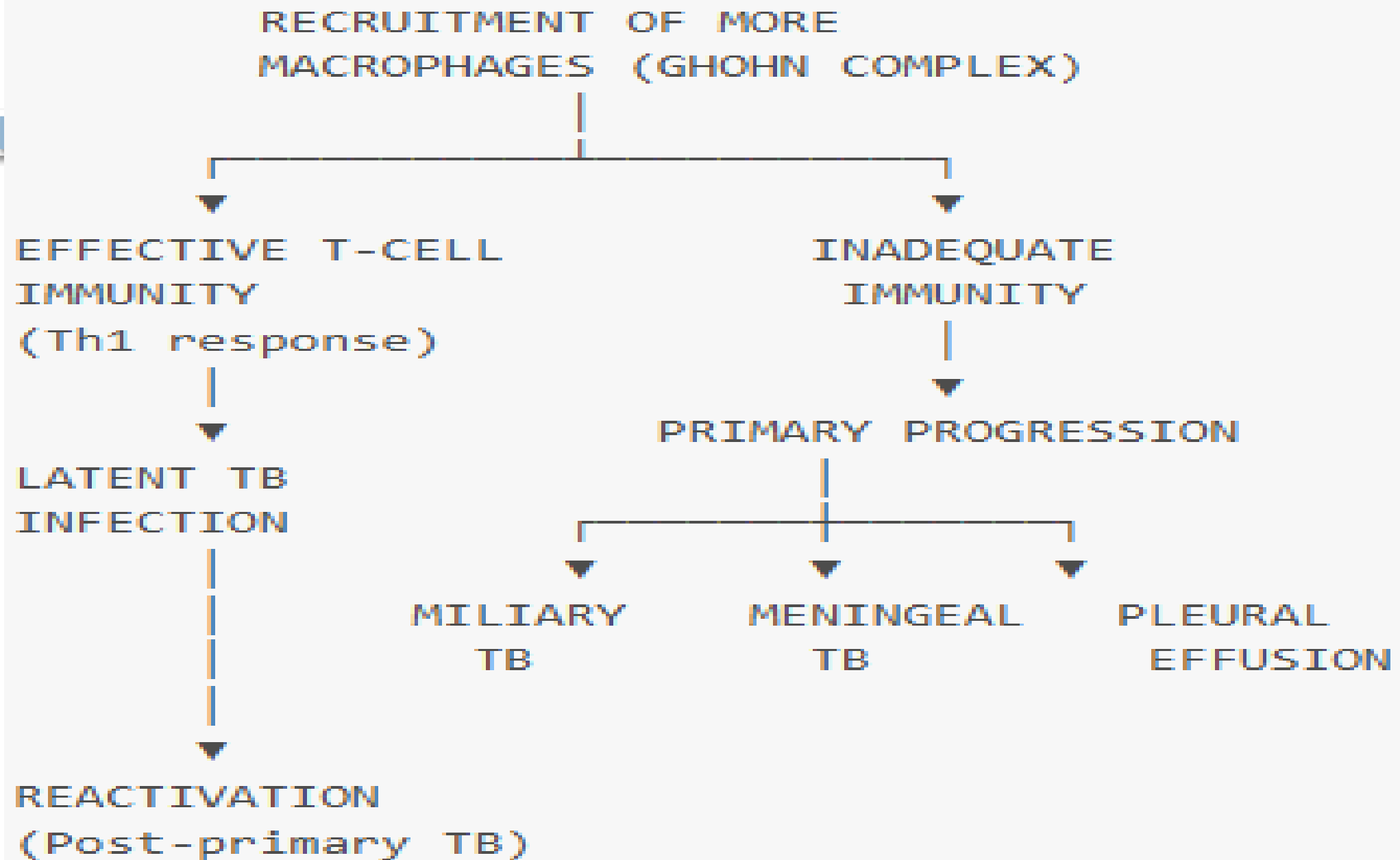
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- Most adults no longer transmit the organism:
 - **2 wks** after beginning chemotherapy.
 - **Some** patients remain infectious for many weeks.

- Young children with tb. rarely infect other children/adults.

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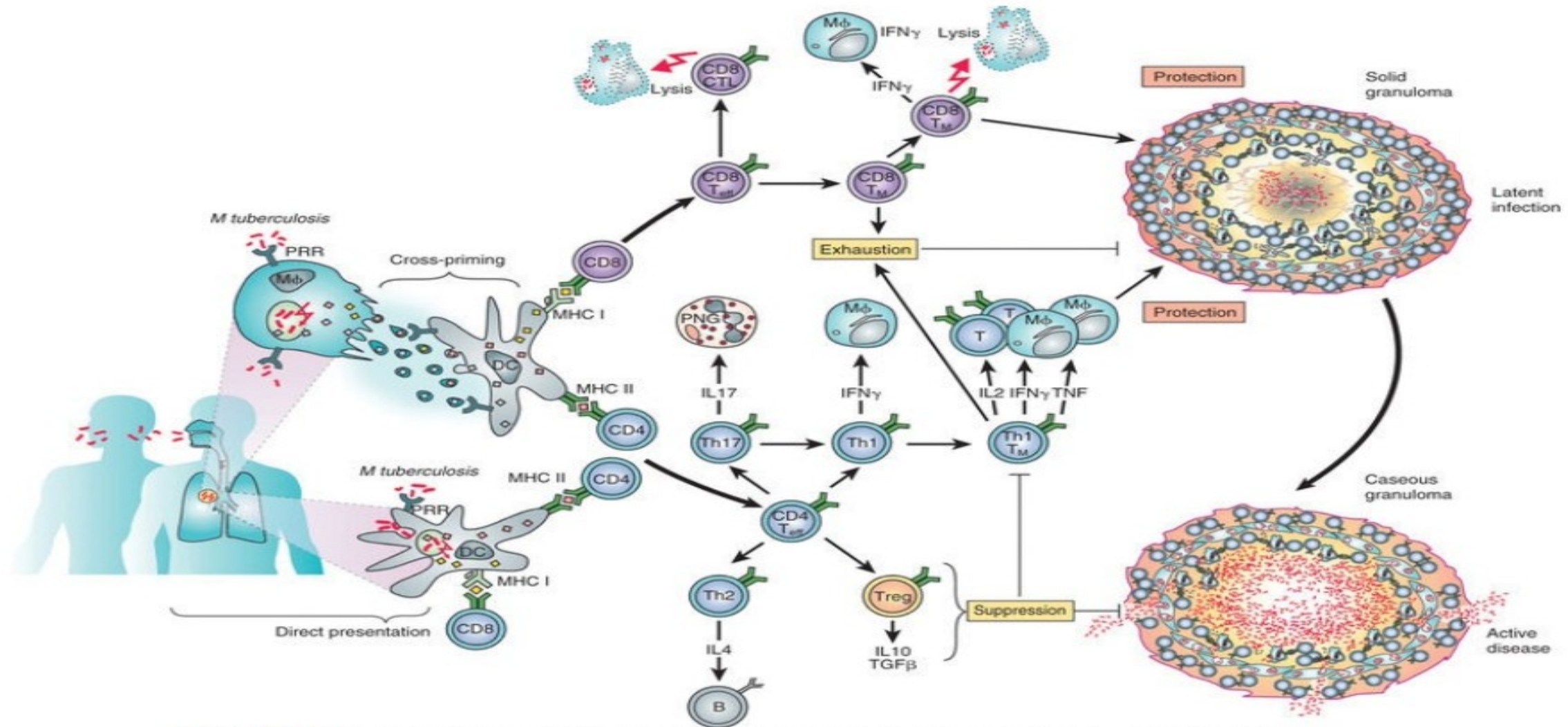
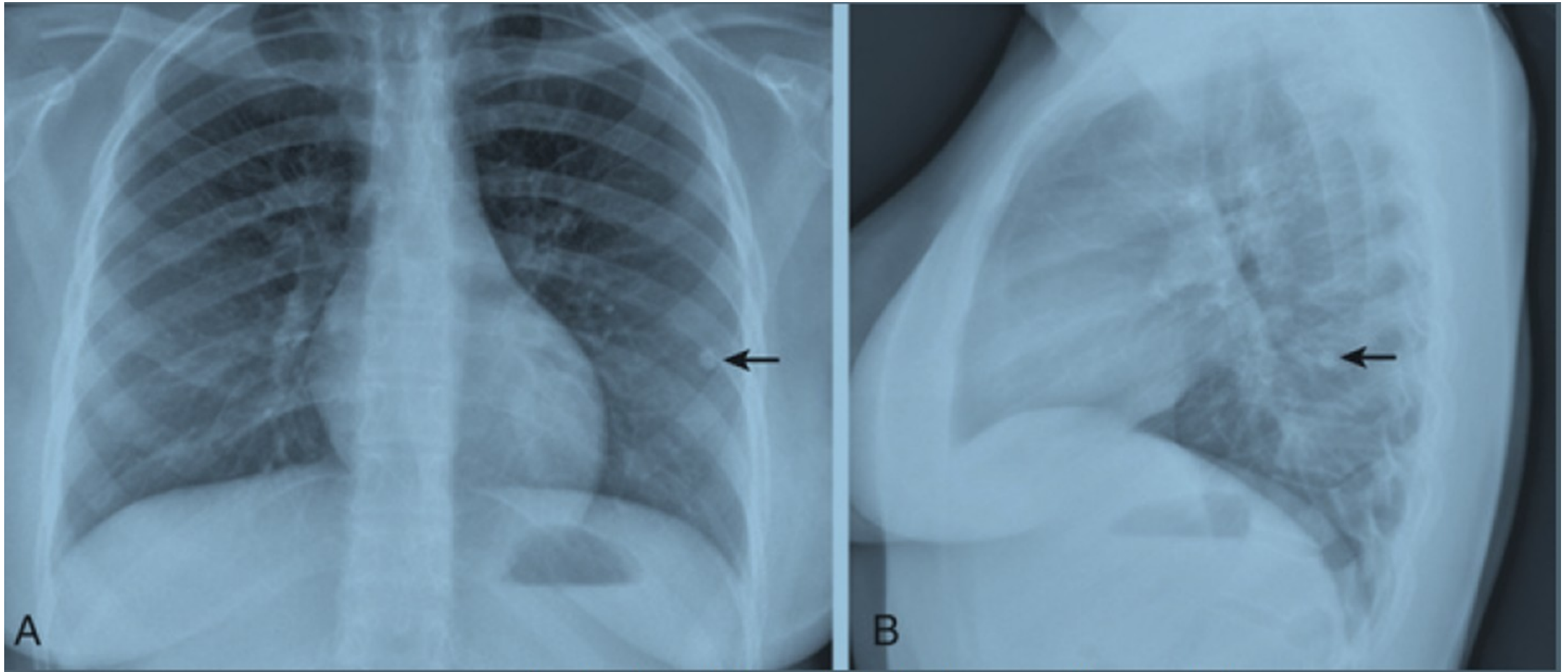


FIG. 242.10 Overview of the immune response in tuberculosis. Control of *Mycobacterium tuberculosis* is mainly the result of productive teamwork between T-cell populations and macrophages (Mφ). *M. tuberculosis* survives within macrophages and dendritic cells (DCs) inside the phagosomal compartment. Gene products of major histocompatibility complex (MHC) class II are loaded with mycobacterial peptides that are presented to CD4 T cells. CD8 T-cell stimulation requires loading of MHC I molecules by mycobacterial peptides in the cytosol, either by overexpression of

CXR of healed primary pulmonary TB

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09/13/2026

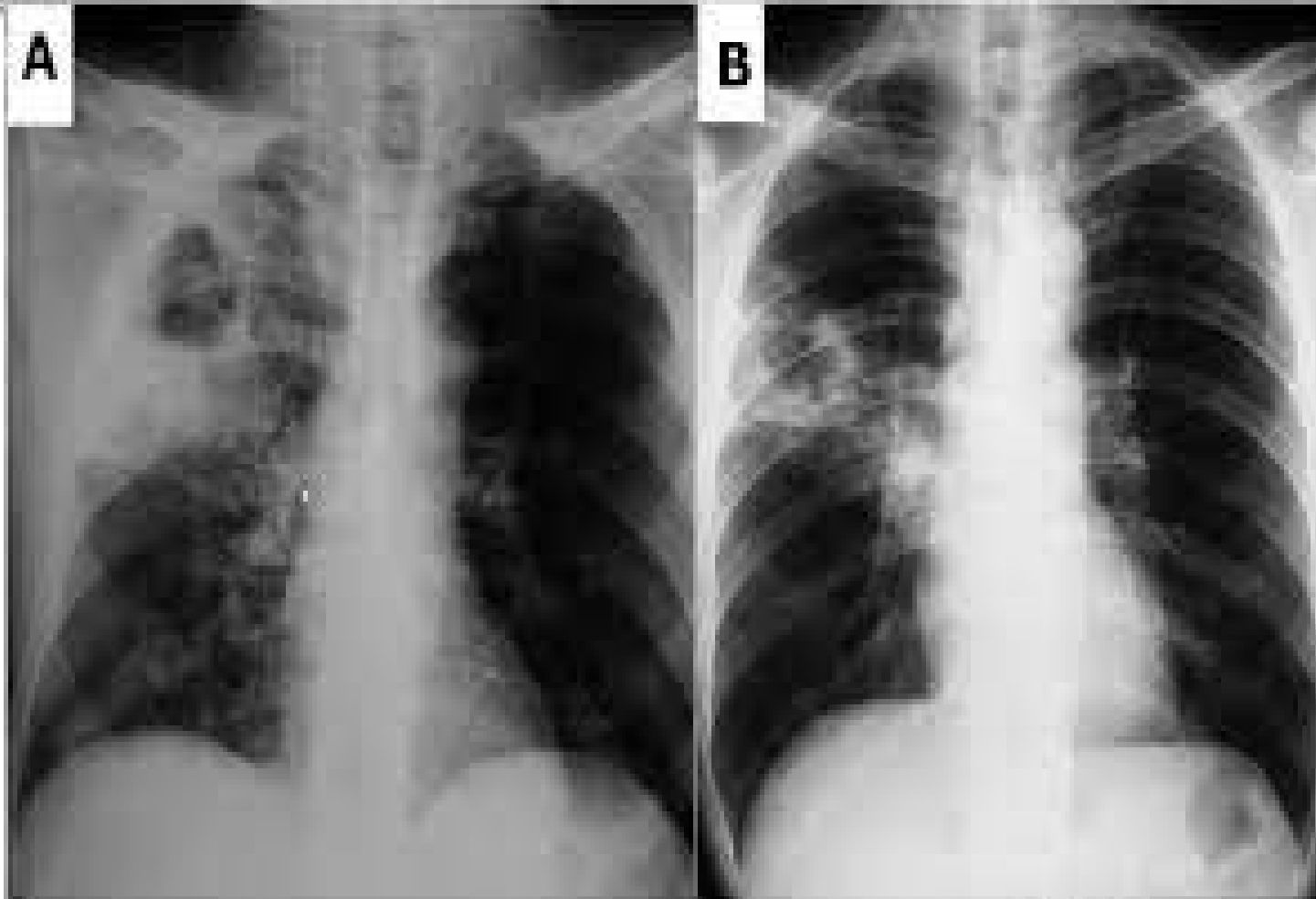




FIG. 242.9 Right-sided hilar lymphadenopathy and collapse-consolidation lesions of primary tuberculosis in a 4 yr old child.

Pathogenesis.....

33

□ **Disseminated tuberculosis occurs if:**

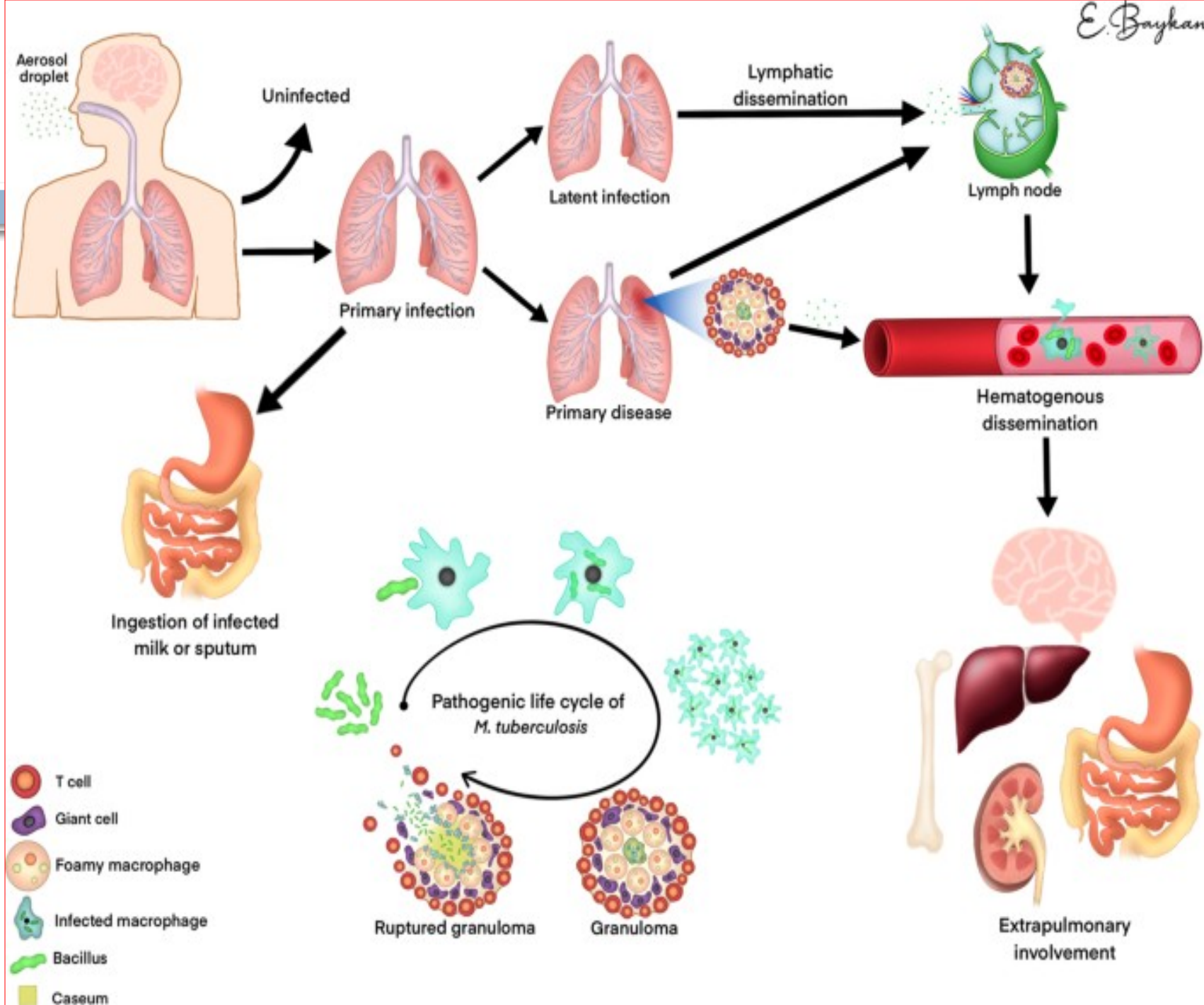
- **Large** number of circulating bacilli &
- **Weak** Host cellular immune response.

Clinical presentation

34

□ **3 major clinical stages:**

- **Exposure**
- **LTBI**
- **Disease**



1. Latent tuberculosis infection (LTBI)

35

- Occurs after inhalation of infective droplet nuclei containing M.tb.
- **Hallmark** of this stage.
 - Reactive TST or IGRA and
 - No clinical & radiographic abnormalities
- Sometimes there might be **granuloma or calcifications**

LTBI..... Natural Hx

36

- Untreated **infants** with LTBI have:
 - Up to a 40% likelihood of developing tuberculosis within 9 mo
 - The risk for progression ↓ing gradually through childhood to adult lifetime rates of 5–10%.
 - For patients with HIV infection, the risk of developing TB is **7-10% per year**.
- The greatest risk for progression occurs in the 1st 2 yr after infection

LTBI.....

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Children at ↑ed risk for progression of LTBI to tb. disease:

➡ diabetes mellitus

➡ chronic renal failure

➡ malnutrition

➡ congenital or acquired immunodeficiencies

LTBI.....

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Risk factors for progression of LTBI to tuberculosis disease Cont.

- ➡ Infants & children ≤ 4 yr of age, especially < 2 yrs
- ➡ Persons who are immunocompromised
- ➡ Persons co-infected with HIV
- ➡ Persons with skin test conversion in the past 1–2 yr

Disease stage forms

39

- Tuberculosis presents in children in various clinical forms:
 - A. Primary pulmonary tuberculosis
 - B. Acute disseminated tuberculosis: meningitis & miliary.
 - C. Post-primary pulmonary tuberculosis
 - D. Extra-pulmonary tuberculosis.

A. Primary pulmonary tuberculosis

40

- Occurs most often in children <5 years of age.
- is **asymptomatic** in the majority of cases .
 - 10% = Symptomatic
 - $\approx 70\%$ = subpleural foci.
- ≥ 2 primary foci = 25% of cases.
- The **hallmark** is
 - Relatively **large size of the regional lymphadenitis** compared with the relatively small size of the initial lung focus.

Complications 1^o PTB

42

- Local complications may occur
 - Collapse consolidation
 - endobronchial tuberculosis
 - Broncho esophageal fistula. Sub carina LN
 - Bronchiectasis Rarely
- Radiograph is more ill than SSx.
 - In 50% of cases

B. Acute disseminated tuberculosis

43

- Early presentation of primary infection
- Occurs within 2–10 mths
- Caused by dissemination of bacilli from the primary infection through the **bloodstream**
- They can occur:
 - At all ages, (most often in ≤ 2 years of age)
 - Not vaccinated with BCG.
- Are very serious & often fatal if late diagnosed & treatment

Tuberculous (TB) meningitis

44

- Accounts 0.3% of untreated **TBIs** in children.
- Often occurs in children **6mo - 4yrs**
- **Brain stem** is common site of involvement and CN III ,VI and VII are affected
- **Communicating hydrocephalus**
- Start initially with non specific complaint ; irritability headache or vomiting.
- Later on **Changes in state of consciousness ,Strabismus, neck rigidity, coma**

TB meningitis ... stages

45

□ **Stage one:**

- Is characterized by nonspecific Sxs: fever, headache, irritability, drowsiness & malaise.
- Focal neurologic signs are absent

□ **Stage two**

- Lethargy, nuchal rigidity, seizures
- Positive Kernig or Brudzinski signs, hypertonia
- Vomiting, cranial nerve palsies & other focal neurologic signs.

□ **Stage three:**

- Is marked by coma
- Hemiplegia /paraplegia, Decerebrate/decorticate posturing

TB meningitis ...laboratory

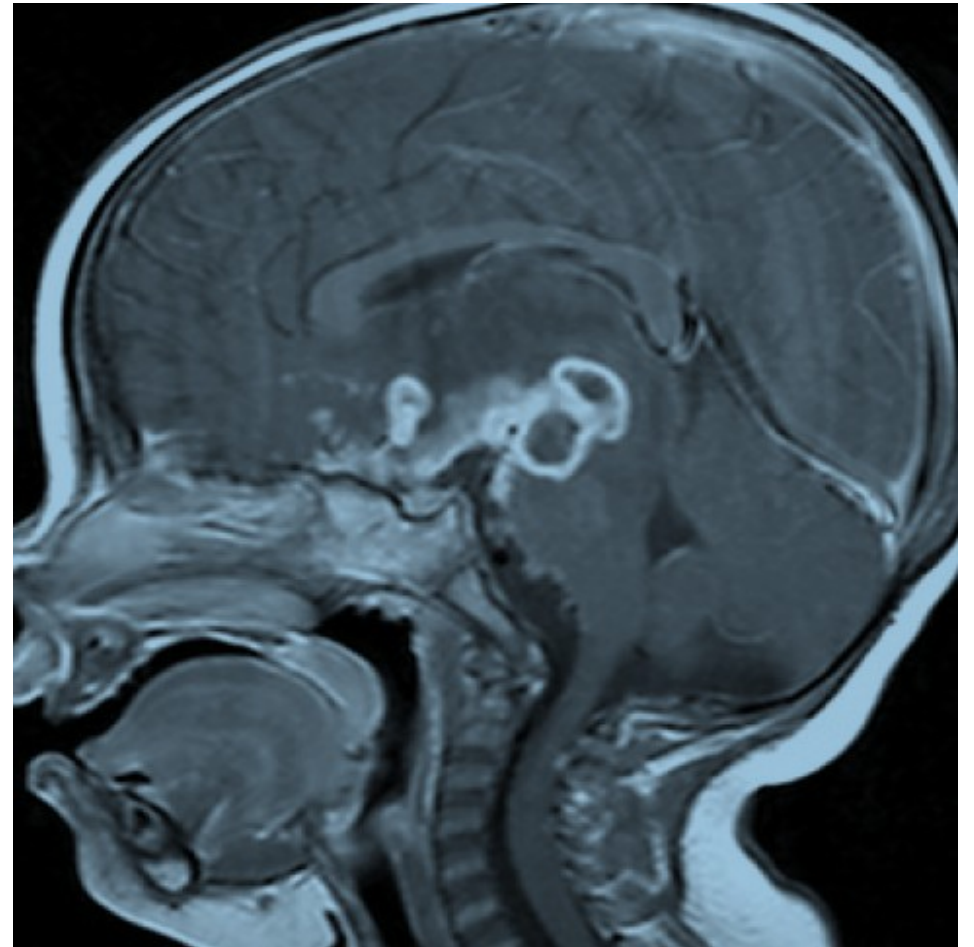
46

- **TST=50, CXR=20-50 ,AFB smear=30 ,CULTURE 50-70**
- **CSF analysis**
 - **Appearance:** Clear, slightly turbid (may form a "cobweb" clot on standing).
 - **Cells:** 10-500 cells/μL (predominantly lymphocytes).
 - **Protein:** Elevated (1-5 g/L).
 - **Glucose:** Low (< 40% of serum glucose).
 - **GeneXpert:** Sensitivity is moderate but recommended.
 - [..\cobweb.mp4](#)

Tuberculoma

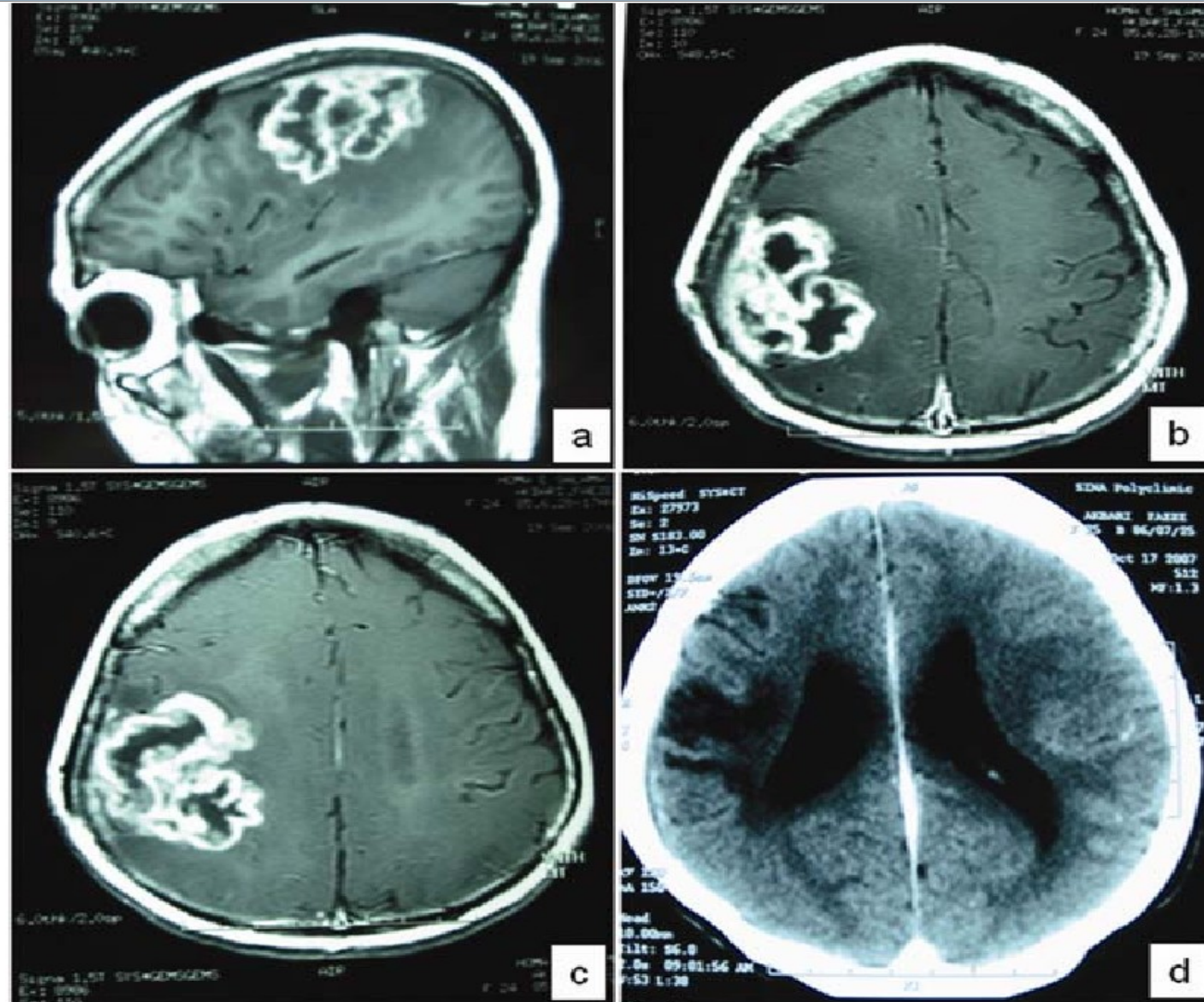
47

- **Infra-tentorial** more common in pediatrics
- Thalamus and cortex
- most often single lesion but may be multiple.
- Headache, fever ,FND and seizure
- **TST** = usually reactive
- **CXR** = usually normal



Tuberculoma in adult

48



showing dura-based contrast-enhancing lesion (a-c). One year post-operative CT scan

TB meningitis ... Prognosis

49

□ **Stage of the illness**

- Patients in the **1st stage** have an excellent outcome
- Most patients in the **3rd stage** who survive have permanent disabilities
 - blindness, deafness,
 - Paraplegia ,
 - Diabetes insipidus
 - Mental retardation.

□ **Age of the patient**

- The prognosis for young infants is generally worse than for older children

□ **Time of initiation of Rx**

Basilar enhancement



Acute miliary tuberculosis

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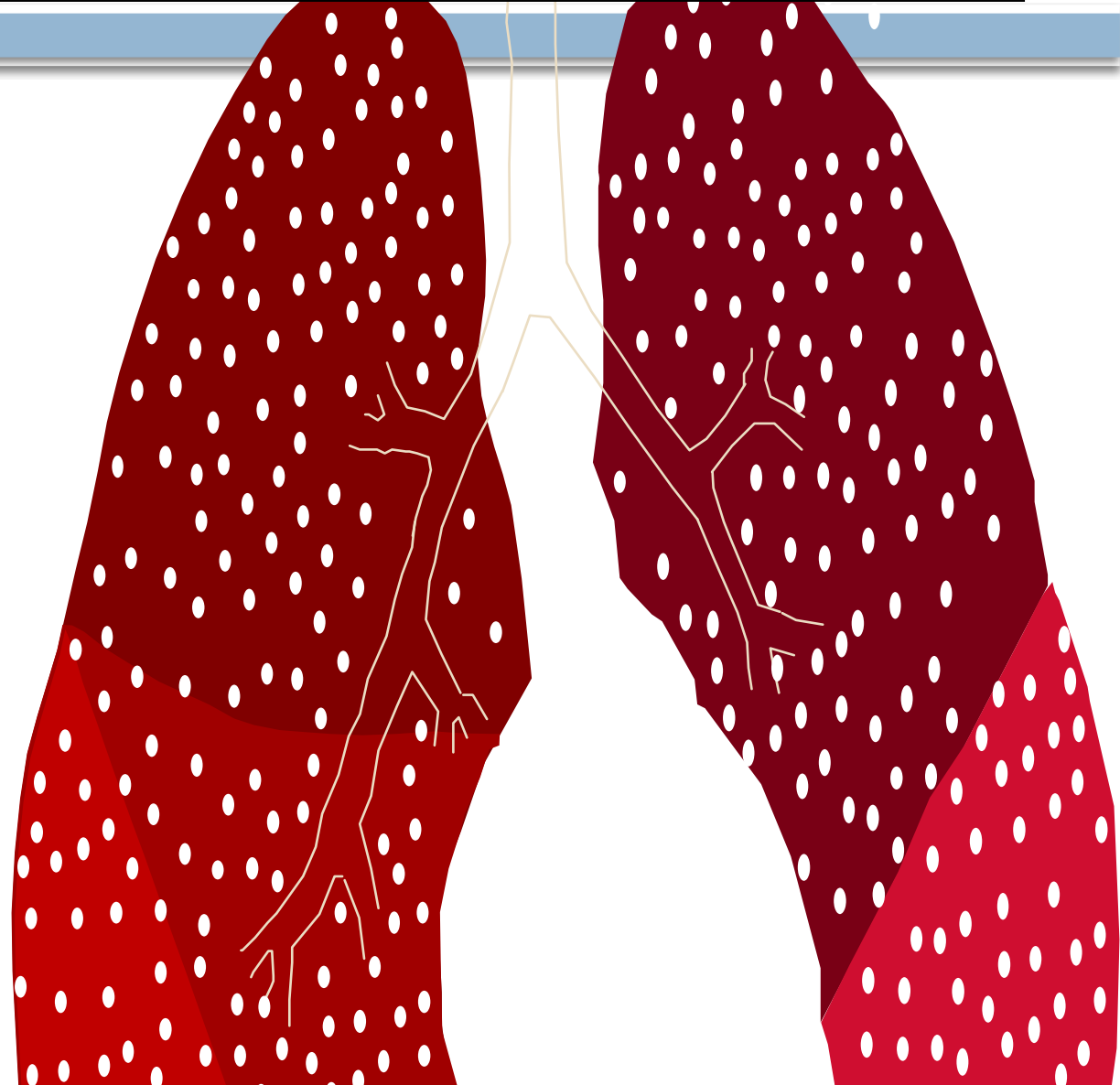
- Generalized dissemination through bloodstream
 - Caseous focus ruptures into blood vessel
- Usually occurs in the 2-6 months after primary infection
 - ❖ **Can** occurs within the first weeks after primary infection
- It is a severe condition with high fever at 39–40°C
- Fatal if no treatment

Miliary TB

53

- Is common in late HIV/AIDS disease.
- Generalized lymphadenopathy and Hepatosplenomegaly are present in **>50%** of cases
- TB meningitis or peritonitis. 20-40%
- Choroid tubercle.....13-87%
- Characteristic spotted shadows on X Ray.

MILIARY TB





MILIARY TB

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- The Dx rests on strong clinical suspicion
- If Rx is delayed, the prognosis will be badly affected.

Miliary TB

57

- The resolution of is slow.
- **Fever** usually declines within 2-3 weeks of Rx,
- **Steroids**,
 - Air block,
 - Peritonitis , or
 - meningitis is present.
- **The prognosis** is excellent with early diagnosis and adequate chemotherapy.

C. Post-primary pulmonary tuberculosis:-

58

- A delayed result of primary infection
- Usually occurs in adults but may appear in
 - Older children & adolescents
 - > 7 years of age
 - Particularly in the presence of malnutrition or other immunosuppressant.

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D. Extra-pulmonary tuberculosis in children

Tuberculous lymphadenitis/ Scrofula

60

- The most common form.
- Most cases occur within 6-9 mo of initial infection
- Disease is most often **unilateral** , posterior cervical > ant cervical
- Asymptomatic mostly
- TST (70%), **CXR (30%), culture (50%)**
- **Usually, Dx by FNAC**
- Lymph node tuberculosis can resolve if left untreated
- responds well to anti-TB,
- **although the lymph nodes do not return to normal size for months or even years.**

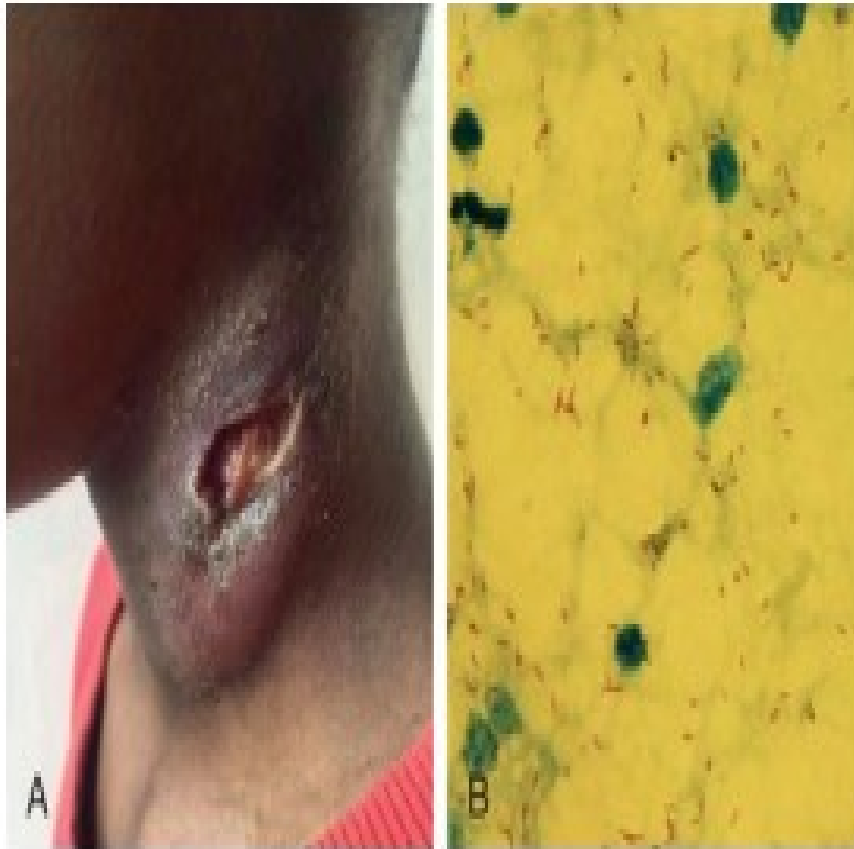


FIG. 242.14 Scrofula. A, Ulcerative lesion 3.2 × 2.1 cm with undermined edges and necrotic base with surrounding induration. B, Acid-fast bacilli. (From Sharawat IK. Scrofula, *J Pediatr* 189:236, 2017).



FIG. 242.15 Scrofula. Tuberculous lymphadenitis with fistula in 4 yr old boy associated with scrofuloderma (arrows). (From Pereira C, Cascais M, Felix M, Salgado M: Scrofula in a child, *J Pediatr* 189:235, 2017).

Tuberculosis of the spine :

62

- Is the 2nd most common
- May occur within the 1st few yrs
- The classic manifestation is
 - Gibbus deformity and
 - Pott disease
 - Pott's Paraplegia
- **Gold standard :- MRI**
- **Surgery** is not necessary for cure, and
- **Prognosis** is excellent with adequate medical Rx



FIG. 242.18 Tuberculosis of the spine in a toddler. (From Feder Jr HM, Rigos L, Teti K: Pott's disease in a Connecticut toddler, *Lancet* 388:504–

Pleural TB

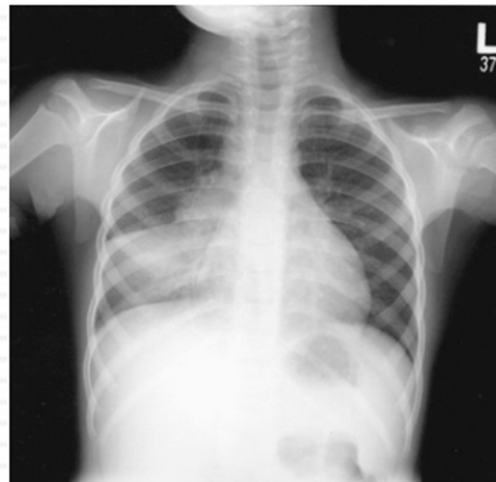
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- ~**2–5%** of all TB cases
- Uncommon(<6 yr), rare (<2 yr)
- **Lab**
 - TST (70–80%),
 - Cultures of the fluid (<30%)
 - GeneXpert... 20-30
 - **ADA** > 40 U/L: Strongly suggestive of Tuberculosis.
- **Prognosis**
 - Excellent
 - Radiographic resolution often takes months.

Medical History

65

Diagnosis of Paediatric TB



Why is Pediatric TB Diagnosis Difficult?

66

- Paucibacillary Disease:
- Difficulty in Sample Collection:
- Non-Specific Symptoms:
- Low Sensitivity of Tests:

For fair degree Dx of childhood

67

TB :

laboratory

- Tuberculin skin test (TST) / IGRA
- Chest radiograph
- Sputum microscopy/ NAAT
- HIV testing.

Medical History

- contact with a smear-+ve
- SSxs complex of TB

Clinical manifestation: S_x of childhood TB

68

- **Chronic cough:**

- Is a persistent cough
- Present for >2 wks & that is not improving.

- **Fever:**

- Fever of >38°C for 14 days
- After common causes like malaria, pneumonia have been excluded.

- **Weight loss:**

- Documented weight loss ,Failure to gain weight,
- Even after being treated in a nutritional Rx.

PHYSICAL SIGNS REQUIRING I_x FOR TB:

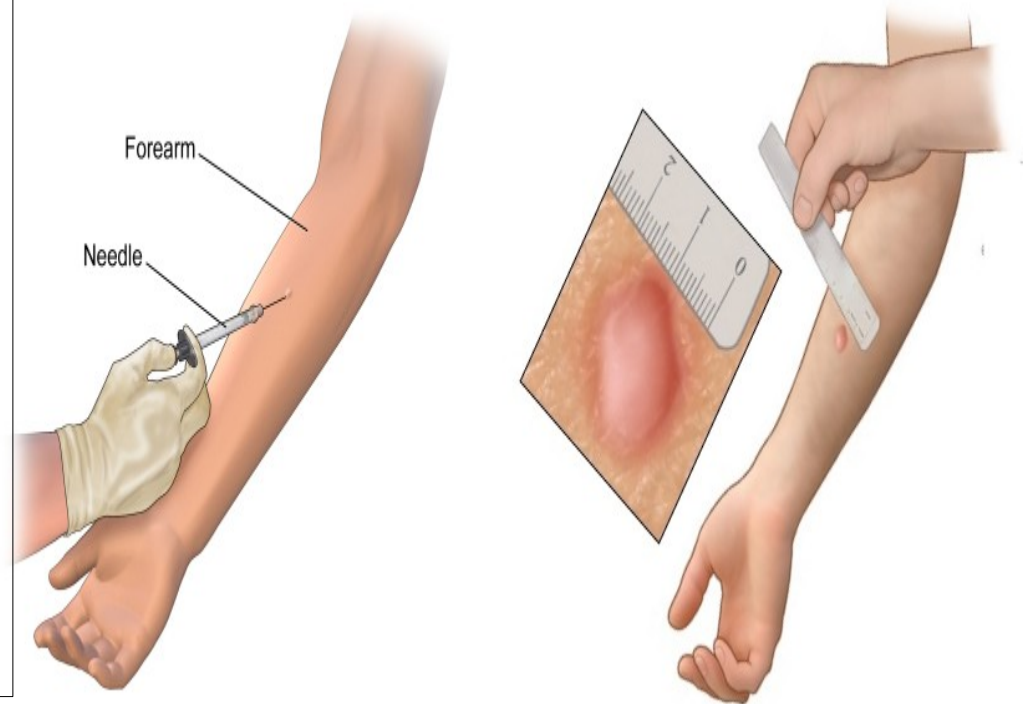
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1. Meningitis not responding to Rx with antibiotics, with sub-acute onset
2. Pleural effusion or pericardial effusion
3. Distended abdomen with ascites
4. Non-painful enlarged joints

Mantoux Test/ TST

70

- **Technique**:- 0.1ml (5 TU) of PPD injected
- **Reading**: induration (not erythema) at 48–72 hours



TST Recommendations

71

Children for whom immediate TST is indicated

- ➡ Contacts of people with confirmed or suspected contagious TB
- ➡ Children with radiographic or clinical findings suggesting TB disease
- ➡ Children immigrating from countries with endemic infection

Children who should have annual TST:

- Children infected with HIV
- Incarcerated adolescents

Definitions of Positive TST Results

72

INDURATION ≥ 5 MM

- ➡ Close contact with known/suspected contagious people.
- ➡ Children suspected to have tuberculosis disease:
 - Findings on CXR consistent with active or previously tuberculosis disease
 - Clinical evidence of tuberculosis disease
- ➡ Children receiving immunosuppressive therapy or with immunosuppressive conditions, including HIV infection

Definitions of Positive.....

73

INDURATION ≥ 10 MM

➡ **Children at ↑ed risk of disseminated tuberculosis disease:**

- Children <4 yr of age
- Children with medical conditions: lymphoma, DM, CRF
- Malnutrition

➡ **Children with ↑ed exposure to tuberculosis disease:**

- Children born in high-prevalence regions of the world

Definitions of Positive TST.....

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INDURATION ≥ 15 MM

Children 4 yr of age or older without any risk factors.

Drug-resistant TB

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- is a laboratory diagnosis.

RISK FACTORS FOR DRUG-RESISTANT TUBERCULOSIS

- ❖ **Personal or contact history of treatment for tuberculosis**
- ❖ **Contacts of patients with drug-resistant tuberculosis**
- ❖ **Birth or residence in a country with a high rate of drug resistance**
- ❖ **Poor response to standard therapy**
- ❖ **Positive sputum smears (acid-fast bacilli) or culture ≥ 2 mo after initiating appropriate therapy**

□ **Primary Resistance**

- Contacted MDR patient

□ **Secondary resistance**

- occurs when drug resistant emerge during treatment.

Drug-resistant TB

81

□ **Mono-resistant TB**

- Resistant in vitro to one first-line anti-TB drug.

□ **MDR:**

- Resistant to at least both INH & RIF.

□ **Poly-resistant TB:**

- Resistant in vitro to >1 first-line drug,
- **other than** both INH & RIF combination

Drug-resistant TB.....

82

□ **Extensive-drug resistant TB (XDR):**

- MDR-TB

PLUS:

- Resistant to any Fluoroquinolone &

PLUS:

- At least one of 3 following injectables used in anti-TB Rx:
 - Capreomycin
 - kanamycin
 - Amikacin.

Principles of tuberculosis treatment.....

83

1. Chemotherapy
2. Nutritional rehabilitation
3. Health education
4. Family screening
5. Follow up

Commonly Used Drugs for Treatment of Tuberculosis...

84

DRUG	DOSAGE FORMS	DAILY DOSAGE (mg/kg)	TWICE-WEEKLY DOSAGE (mg/kg/dose)	MAXIMUM DOSE	ADVERSE REACTIONS
Ethambutol	Tablets: 100 mg 400 mg	20	50	2.5 g	Optic neuritis (usually reversible), decreased red-green color discrimination, gastrointestinal tract disturbances, hypersensitivity
Isoniazid*	Scored tablets: 100 mg 300 mg Syrup: 10 mg/mL	10-15 †	20-30	Daily: 300 mg Twice weekly: 900 mg	Mild hepatic enzyme elevation, hepatitis, † peripheral neuritis, hypersensitivity
Pyrazinamide*	Scored tablets: 500 mg	30-40	50	2 g	Hepatotoxic effects, hyperuricemia, arthralgias, gastrointestinal tract upset
Rifampin*	Capsules: 150 mg 300 mg Syrup formulated from capsules	15-20	15-20	600 mg	Orange discoloration of secretions or urine, staining of contact lenses, vomiting, hepatitis, influenza-like reaction, thrombocytopenia, pruritus Oral contraceptives may be ineffective.

First-Line Anti-TB Drugs

85

- **Isoniazid (INH or H):** Bactericidal.
- **Rifampicin (RIF or R):** Bactericidal, sterilizing.
- **Pyrazinamide (PZA or Z):** Sterilizing (active in acidic environment).
- **Ethambutol (EMB or E):** Bacteriostatic (prevents resistance).

Treatment of tuberculosis in children

86

- Should be Rxed with
 - 4-drug initial phase.
 - 2-drug for maintenance
- **EPTB** can be Rxed with the same regimens as pul.disease.
 - Exceptions are the
 - Disseminated TB disease
 - Meningitis
 - *Bone and joint,*
 - ➔for which the recommended duration is 9 to 12 months.
- **HIV positive**
 - 9month ???

4-month regimen

87

- children age 0-16years(≥ 3 kg)
- Non-severe ,
- Smear negative,
- Presumed drug-susceptible TB

Treatment of tuberculosis in children.....

88

□ **Pyridoxine :**

- For all
- **Routine supplementation:** ~0.5–1 mg/kg/day
- **Increase dose if neuropathy develops:** may be increased (per symptom severity).

Strong Indications for treatment with steroids

89

- **TB meningitis**
 - ↓ mortality and neurologic sequelae
- **TB pericarditis**
 - ↓ risk of constrictive pericarditis
- **Miliary tuberculosis**
 - Especially with hypoxemia or severe systemic illness
- **Common choices:**
 - **Dexamethasone** (TB meningitis)
 - **Prednisolone** (most other indications)

END TB strategy

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Vision:

A world free of TB

*Zero TB deaths,
Zero TB disease, and
Zero TB suffering*

Goal:

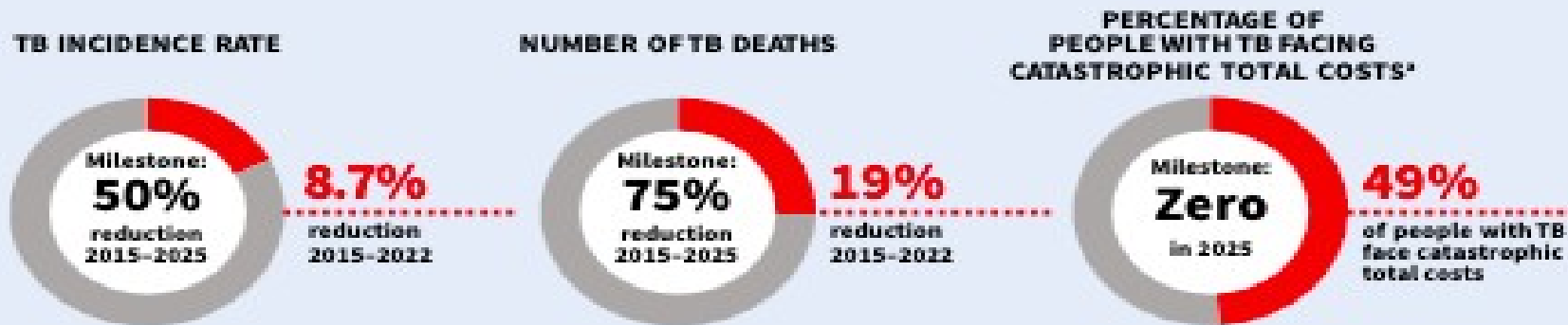
End the Global TB epidemic

	MILESTONES		TARGETS	
	2020	2025	SDG* 2030	END TB 2035
<i>Reduction in number of TB deaths compared with 2015 (%)</i>	35%	75%	90%	95%
<i>Reduction in TB incidence rate compared with 2015 (%)</i>	20%	50%	80%	90%
<i>TB-affected families facing catastrophic costs due to TB (%)</i>	0%	0%	0%	0%

END TB strategy

91

WHO End TB Strategy: 2025 milestones



2018 UN high-level meeting on TB: treatment targets



Reference

92

- Nelson 22st ed
- National TB, Leprosy, Tb/HIV participant manual 2022
- WHO 2024/25 TB report

THANK YOU!!!

THE

END

TB

STRATEGY